

 ECOSYSTEM SHOWCASE • HUAWEI HARMONYOS NEXT

Intelligent Accessible Navigation on HarmonyOS

Native architectural barrier awareness powered by ArkTS, low-latency Huawei Kits, and WCAG AAA accessibility.

 Designed by the **Brainstorm Team** for Huawei Devices

Tailored Mobility Intelligence

"Generally accessible" does not mean accessible for every individual mobility profile.



Stairs & Curbs

Accurate step counts.



Lifts & Ramps

Clearance and reliability.



Doors & Thresholds

Opening width and weight.



Surface Types

Incline and pavement quality.

Engineered for Huawei Device Users

Individual Need Profiles

Custom routing for manual wheelchairs, powerchairs, and strollers.

Huawei 1+8+N Synergy

Seamless synchronization across smartphones, tablets, and smartwatches.

Verified Data Provenance

Audited municipal datasets combined with active community reporting.

600 × 400

End-to-End HarmonyOS Navigation Flow

Continuous hardware-accelerated assistance throughout trip planning and execution.

1

Profile Select

Configure custom mobility parameters and barrier filters.

2

GNSS Location

Hardware positioning powered directly by LocationKit.

3

Route Clearance

Evaluate stairs, elevator widths, and slope angles.

4

Live Direction

Real-time orientation via SensorServiceKit haptics.

Data Transparency & Confidence Tracking

Profile: Manual Wheelchair

Accessible Route

- ✓ **Entrance:** Level zero-step access.
- ✓ **Elevator:** 95 cm door clearance.
- ✓ **Corridors:** Over 130 cm width.
- ⚠ **Restroom:** Heavy manual door handle.

Provenance Metadata

Primary Source:	Facility Operator + Field Audits
Last Verified:	October 2026
Confidence Level:	High (On-Site Verified)
Unverified Areas:	Explicit Warning Displayed

Hybrid HarmonyOS NEXT Container

Native ArkUI / ArkTS Layer (ReactApp.ets)

@kit.ArkWeb container hosting offline bundle (\$rawfile) in a secure sandbox.

OpenHarmony API 24

↕ Two-Way Native Bridge (.javaScriptProxy <-> window.HarmonyBridge)

Web / React Runtime (harmony.ts & MapView)

Expo Web bundle, accessible navigation UI, and tactile map routing.

React Native Web

Deep Huawei Hardware Integration

Direct system API hooks into HarmonyOS NEXT low-latency native kits.



@kit.SensorServiceKit

Hardware orientation angle and tactile vibration haptics.



@kit.LocationKit

Energy-efficient GNSS positioning without web overhead.



@kit.BasicServicesKit

Continuous battery SoC monitoring and power thrift mode.



@kit.AbilityKit

System font size multiplier and OS dark mode synchronization.

Digital Accessibility & Build Integrity

Huawei Runtime Stack

- ✓ **DevEco Studio:** Automated hvirow assembleHap build pipeline.
- ✓ **Offline Packaging:** Zero network latency via local \$rawfile bundle.
- ✓ **Error Resilience:** Integrated harmony-boot-error visualizer.

Accessibility (WCAG 2.1)

- ✓ **High Contrast:** Full adherence to strict WCAG AAA contrast ratio.
- ✓ **Screen Reader:** Semantic maneuver announcements for every turn.
- ✓ **Keyboard Traversal:** Unobstructed focus indicator management.

Distribution via Huawei AppGallery

Synergy between Huawei consumer devices and intelligent municipal services.

AppGallery Launch

Streamlined deployment of the native HAP container.

Multi-City Expansion

Scalable data schemas ready for global metropolitan hubs.

Cloud Redundancy

High availability data routing with enterprise resilience.

 SUMMARY & DEPLOYMENT READINESS

Verified HAP Prototype for HarmonyOS NEXT

Validated bidirectional ArkTS bridging, native hardware sensor telemetry, and uncompromising WCAG AAA digital accessibility.

MILESTONE STATUS

Functional HAP Built

ARCHITECTURE

**ArkWeb + ArkTS
Bridge**

NEXT STEP

**Huawei AppGallery
Release**